

specific observation

```
graph TD; A[specific observation] --> B[common]; A --> C[exceptional];
```

common

exceptional

specific observation

```
graph TD; A[specific observation] --> B[common]; A --> C[exceptional]; D["number of standard deviations removed from the mean"] --> B; D --> C;
```

common

exceptional

number of
standard deviations
removed from the mean

specific observation

```
graph TD; A[specific observation] --> B[common]; A --> C[exceptional]; D["number of standard deviations removed from the mean"] --> B; D --> C; E[Z-SCORE] --> D;
```

common

exceptional

number of
standard deviations
removed from the mean

Z-SCORE

$$\bar{x} = 15$$

TEAM 1

$$s = 2.5$$



$$\bar{x} = 15$$

TEAM 1

$$s = 2.5$$



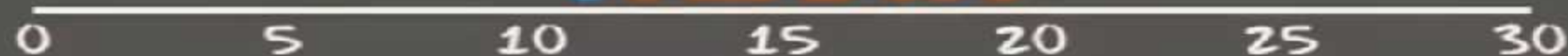
Z-SCORE

$$Z = \frac{x - \bar{x}}{s}$$

$$\bar{x} = 15$$

$$s = 2.5$$

10.8 TEAM 1



Z-SCORE

$$Z = \frac{x - \bar{x}}{s}$$

$$\bar{x} = 15$$

TEAM 1

$$s = 2.5$$

10.8



0 5 10 15 20 25 30

Z-SCORE

$$Z = \frac{10.8 - \bar{x}}{s}$$



Z-SCORE

$$Z = \frac{10.8 - 15}{S}$$

$$\bar{x} = 15$$

10.8 TEAM 1

$$S = 2.5$$

0 5 10 15 20 25 30

Z-SCORE

$$Z = \frac{10.8 - 15}{2.5}$$

$$\bar{x} = 15$$

TEAM 1

$$s = 2.5$$

10.8



Z-SCORE

$$Z = \frac{10.8 - 15}{2.5} = -1.68$$

	Percentage of body covered with tattoos
Player 1	10,8
Player 2	14,1
Player 3	17,6
Player 4	19,3
Player 5	15,4
Player 6	15,3
Player 7	15
Player 8	17,8
Player 9	13,5
Player 10	12,1
Player 11	14,1

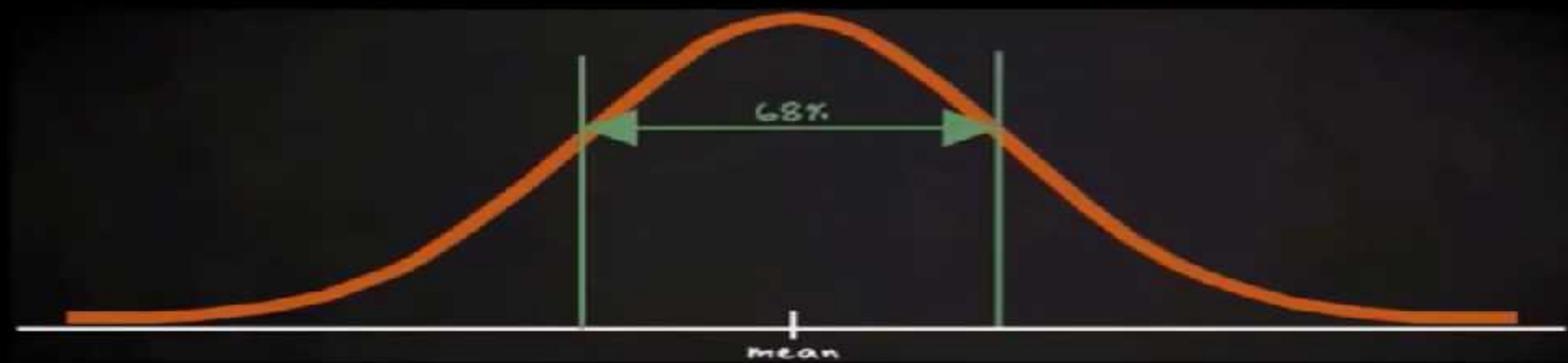
	Percentage of body covered with tattoos	z-score
Player 1	10,8	-1,68
Player 2	14,1	-0,36
Player 3	17,6	1,04
Player 4	19,3	1,72
Player 5	15,4	0,16
Player 6	15,3	0,12
Player 7	15	0
Player 8	17,8	1,12
Player 9	13,5	-0,6
Player 10	12,1	-1,16
Player 11	14,1	-0,36

	Percentage of body covered with tattoos	z-score
Player 1	10,8	-1,68
Player 2	14,1	-0,36
Player 3	17,6	1,04
Player 4	19,3	1,72
Player 5	15,4	0,16
Player 6	15,3	0,12
Player 7	15	0
Player 8	17,8	1,12
Player 9	13,5	-0,6
Player 10	12,1	-1,16
Player 11	14,1	-0,36

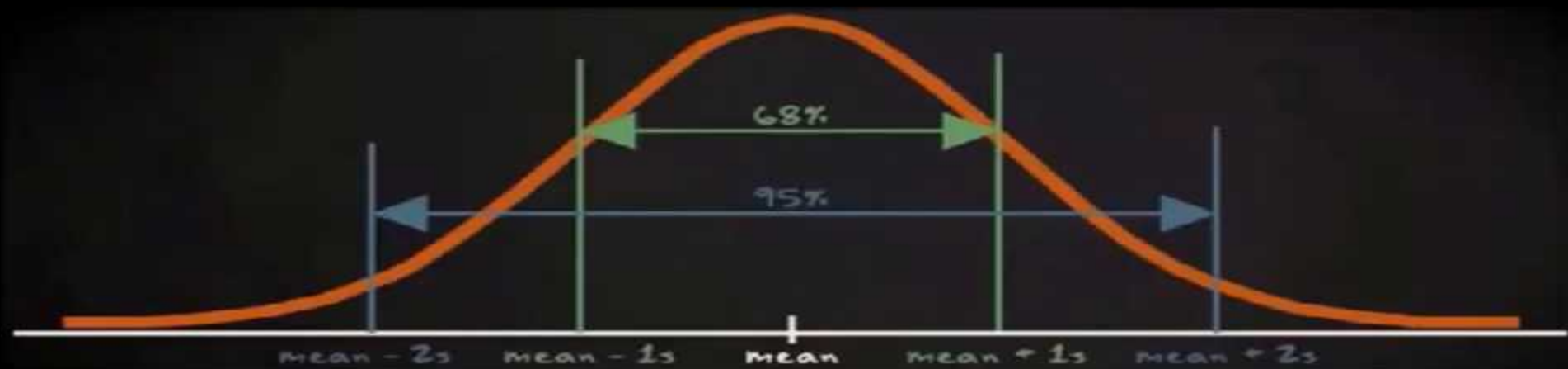
	Percentage of body covered with tattoos	z-score
Player 1	10,8	-1,68
Player 2	14,1	-0,36
Player 3	17,6	1,04
Player 4	19,3	1,72
Player 5	15,4	0,16
Player 6	15,3	0,12
Player 7	15	0
Player 8	17,8	1,12
Player 9	13,5	-0,6
Player 10	12,1	-1,16
Player 11	14,1	-0,36
		0.0

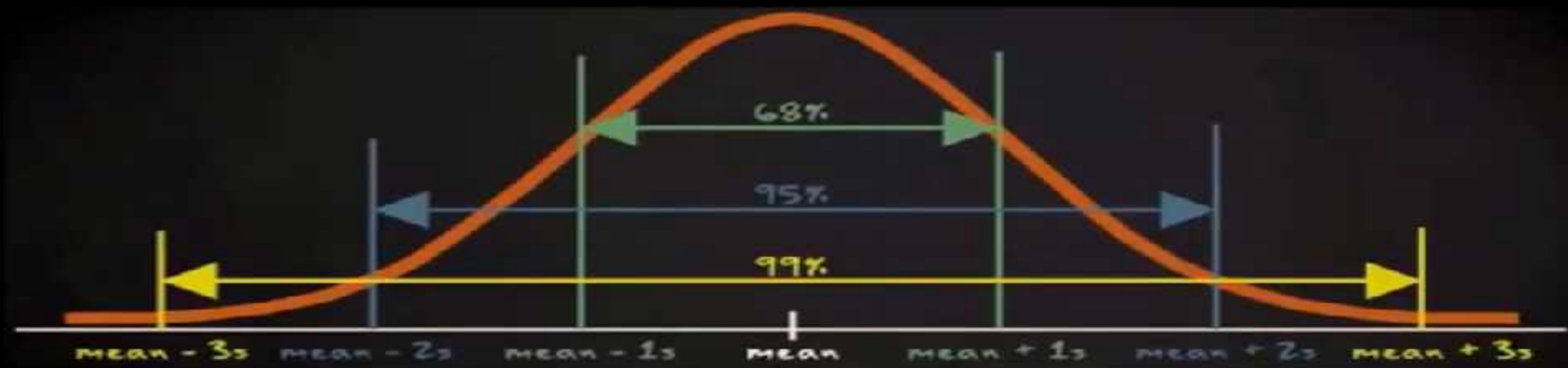












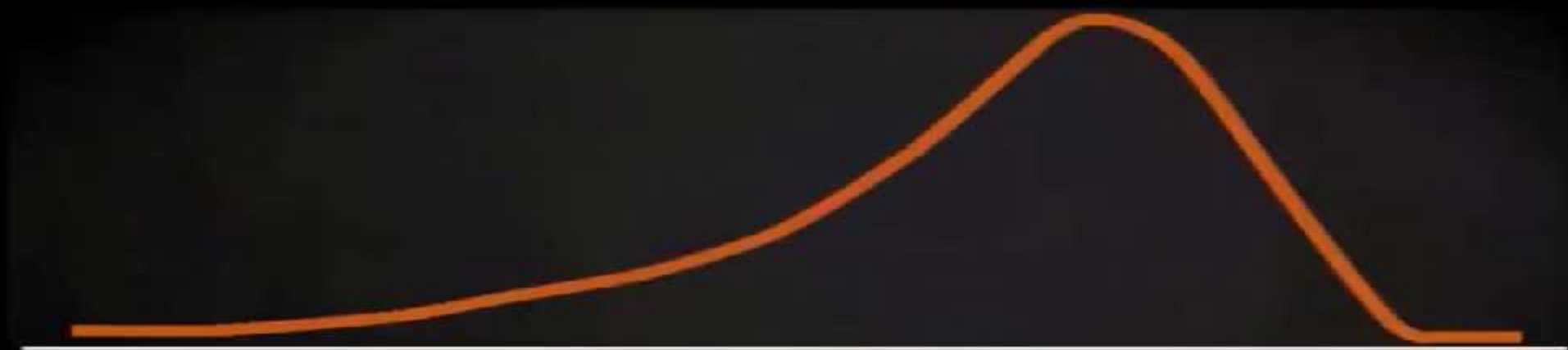






more extreme values

large positive z-scores



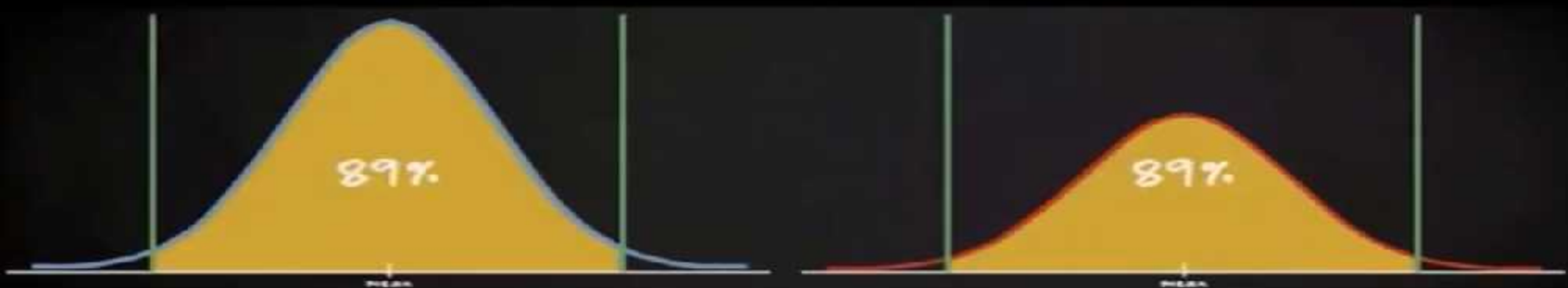
more extreme values

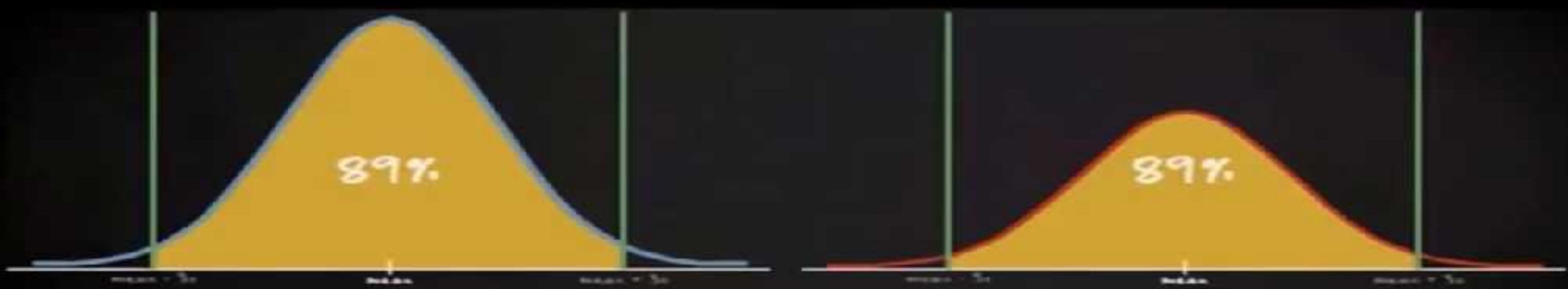


large negative z-scores









TEAM 1
TEAM 2

$$\bar{x} = 15$$
$$S = 2.5$$



$$\bar{x} = 15$$
$$S = 8$$



TEAM 1
TEAM 2

$$\bar{x} = 15$$
$$s = 2.5$$



$$\bar{x} = 15$$
$$s = 8$$



TEAM 2 TEAM 1

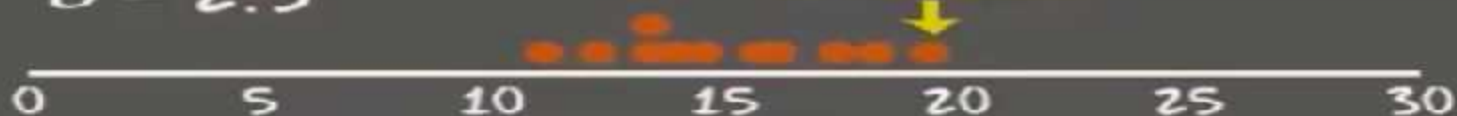


TEAM 1
TEAM 2

$$\bar{x} = 15$$
$$s = 2.5$$

$$z = \frac{17.5 - 15}{2.5} = 1.0$$

not THAT common



$$\bar{x} = 15$$
$$s = 8$$

$$z = \frac{17.5 - 15}{8} = 0.31$$

much more common



recode original
scores into z-scores



STANDARDIZATION

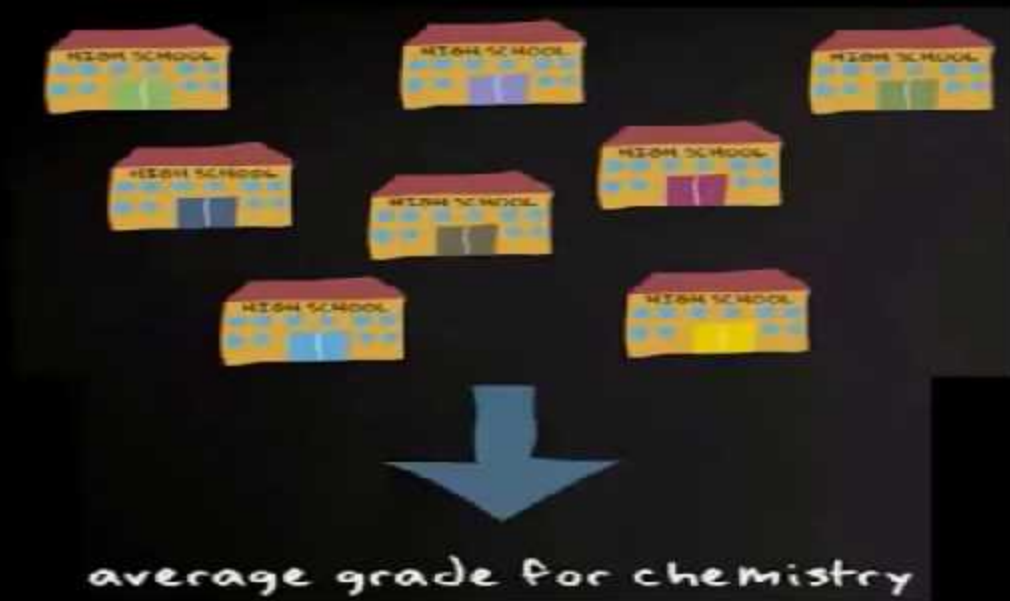
recode original
scores into z-scores



STANDARDIZATION



replace the original scores
by standard deviations
from the mean





average grade for chemistry



average grade for chemistry

0 - 10

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

you want to know:

1. What does the distribution of the variable "average grade for chemistry" look like?

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

you want to know:

2. What is the center of
the distribution?

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

you want to know:

3. The variability
of the distribution

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

you want to know:



4. Construct a box plot

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

you want to know:



5. What is the
z-score of school #3?

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

1. What does the distribution of the variable "average grade for chemistry" looks like?

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

1. What does the distribution of the variable "average grade for chemistry" look like?



quantitative variable

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

1. What does the distribution of the variable "average grade for chemistry" look like?



quantitative variable
small sample

CASES



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

1. What does the distribution of the variable "average grade for chemistry" look like?



quantitative variable
small sample



DOT PLOT

OUTLIER
school #3



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

MODE

MEDIAN

MEAN

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high

~~4,1~~

~~6,2~~

~~6,7~~

7,1

7,4

~~7,4~~

~~7,9~~

~~8,1~~

$$(7,1 + 7,4) / 2 = 7,25$$

MODE

7.4

MEDIAN

MEAN

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high

~~4,1~~

~~6,2~~

~~6,7~~

7,1

7,4

~~7,4~~

~~7,9~~

~~8,1~~



$$(7,1 + 7,4) / 2 = 7,25$$

MODE

7.4

MEDIAN

7.25

MEAN

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

$$\bar{x} = \frac{\sum x}{n}$$

MODE

7.4

MEDIAN

7.25

MEAN

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

54.9

MODE

7.4

MEDIAN

7.25

MEAN

$$\bar{x} = \frac{\sum x}{n}$$

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7
	<u>54.9</u>

MODE

7.4

MEDIAN

7.25

MEAN

$$\bar{x} = \frac{54.9}{n}$$

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7
	<u>54.9</u>

$$\bar{x} = \frac{54.9}{8}$$

MODE

7.4

MEDIAN

7.25

MEAN

Average grade
chemistry

School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7 +
	<u>54.9</u>

$$\bar{x} = 6.86$$

MODE

7.4

MEDIAN

7.25

MEAN

6.86

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7
	<u>54.9</u>

MODE

7.4

MEDIAN

7.25

MEAN

6.86

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7
	<u>54.9</u>

MODE

7.4

MEDIAN

7.25

MEAN

6.86

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

MEASURES OF DISPERSION

RANGE



HIGHEST VALUE - LOWEST VALUE

8.1-4.1

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

MEASURES OF DISPERSION

RANGE



HIGHEST VALUE - LOWEST VALUE

$$8.1 - 4.1 = 4$$

INTERQUARTILE RANGE



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

MEASURES OF DISPERSION

RANGE



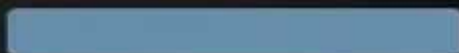
HIGHEST VALUE - LOWEST VALUE

$$8.1 - 4.1 = 4$$

INTERQUARTILE RANGE



$Q3 - Q1$



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high

~~4,1~~

~~6,2~~

~~6,7~~

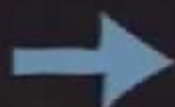
7,1

7,4

~~7,4~~

~~7,9~~

~~8,1~~



$$(7,1 + 7,4) / 2 = 7,25$$

MEASURES OF DISPERSION

RANGE



HIGHEST VALUE - LOWEST VALUE

$$8,1 - 4,1 = 4$$

INTERQUARTILE RANGE



$Q3 - Q1$



	Average grade chemistry	low - high
School 1	7,4	4,1
School 2	7,9	6,2
School 3	4,1	6,7
School 4	8,1	7,1
School 5	6,2	
School 6	7,1	
School 7	7,4	
School 8	6,7	

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q3 - Q1$

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high

~~4,1~~

6,2

6,7

~~8,1~~



$$Q1 = (6,2 + 6,7) / 2 = 6,45$$

MEASURES OF DISPERSION

RANGE



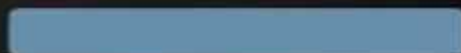
HIGHEST VALUE - LOWEST VALUE

$$8,1 - 4,1 = 4$$

INTERQUARTILE RANGE



$$Q3 - Q1$$



	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high

~~4,1~~

6,2

6,7

~~8,1~~



$$Q1 = (6,2 + 6,7) / 2 = 6,45$$

MEASURES OF DISPERSION

RANGE



HIGHEST VALUE - LOWEST VALUE

$$8,1 - 4,1 = 4$$

INTERQUARTILE RANGE



$Q3 - Q1$

$$6,45$$

	Average grade chemistry	low - high
School 1	7,4	
School 2	7,9	
School 3	4,1	
School 4	8,1	
School 5	6,2	7,4
School 6	7,1	7,4
School 7	7,4	7,9
School 8	6,7	8,1

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE

$$8.1 - 4.1 = 4$$

INTERQUARTILE RANGE

Q3 - Q1

$$6,45$$

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high



~~7,4~~
7,4
7,9
~~8,1~~



$$Q3 = (7,4 + 7,9) / 2 = 7,65$$

MEASURES OF DISPERSION

RANGE



HIGHEST VALUE - LOWEST VALUE

$$8,1 - 4,1 = 4$$

INTERQUARTILE RANGE



Q3 - Q1

$$6,45$$

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high



~~7,4~~
7,4
7,9
~~8,1~~



$$Q3 = (7,4 + 7,9) / 2 = 7,65$$

MEASURES OF DISPERSION

RANGE



HIGHEST VALUE - LOWEST VALUE

$$8,1 - 4,1 = 4$$

INTERQUARTILE RANGE



Q3 - Q1

$$7,65 - 6,45$$

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

low - high



~~7,4~~
7,4
7,9
~~8,1~~



$$Q3 = (7,4 + 7,9) / 2 = 7,65$$

MEASURES OF DISPERSION

RANGE



HIGHEST VALUE - LOWEST VALUE

$$8,1 - 4,1 = 4$$

INTERQUARTILE RANGE



$Q3 - Q1$

$$7,65 - 6,45 = 1,2$$

	Average grade chemistry
School 1	7,4
School 2	7,9
School 3	4,1
School 4	8,1
School 5	6,2
School 6	7,1
School 7	7,4
School 8	6,7

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q3 - Q1$
 $7,65 - 6,45 = 1,2$

STANDARD DEVIATION

	Average grade chemistry	$x - \bar{x}$
School 1	7,4	0,54
School 2	7,9	1,04
School 3	4,1	-2,76
School 4	8,1	1,24
School 5	6,2	-0,66
School 6	7,1	0,24
School 7	7,4	0,54
School 8	6,7	-0,16

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q3 - Q1$
 $7,65 - 6,45 = 1,2$

STANDARD DEVIATION

	Average grade chemistry	$x - \bar{x}$	$(x - \bar{x})^2$
School 1	7,4	0,54	0,2916
School 2	7,9	1,04	1,0816
School 3	4,1	-2,76	7,6176
School 4	8,1	1,24	1,5376
School 5	6,2	-0,66	0,4356
School 6	7,1	0,24	0,0576
School 7	7,4	0,54	0,2916
School 8	6,7	-0,16	0,0256

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q_3 - Q_1$
 $7,65 - 6,45 = 1,2$

STANDARD DEVIATION

	Average grade chemistry	$x - \bar{x}$	$(x - \bar{x})^2$
School 1	7,4	0,54	0,2916
School 2	7,9	1,04	1,0816
School 3	4,1	-2,76	7,6176
School 4	8,1	1,24	1,5376
School 5	6,2	-0,66	0,4356
School 6	7,1	0,24	0,0576
School 7	7,4	0,54	0,2916
School 8	6,7	-0,16	0,0256
			<u>11.3388</u>

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q3 - Q1$
 $7,65 - 6,45 = 1,2$

STANDARD DEVIATION

	Average grade chemistry	$x - \bar{x}$	$(x - \bar{x})^2$
School 1	7,4	0,54	0,2916
School 2	7,9	1,04	1,0816
School 3	4,1	-2,76	7,6176
School 4	8,1	1,24	1,5376
School 5	6,2	-0,66	0,4356
School 6	7,1	0,24	0,0576
School 7	7,4	0,54	0,2916
School 8	6,7	-0,16	0,0256
			<u>11.3388</u>

$$s = \sqrt{\frac{11.3388}{7}}$$

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q3 - Q1$
 $7,65 - 6,45 = 1,2$

STANDARD DEVIATION

	Average grade chemistry	$x - \bar{x}$	$(x - \bar{x})^2$
School 1	7,4	0,54	0,2916
School 2	7,9	1,04	1,0816
School 3	4,1	-2,76	7,6176
School 4	8,1	1,24	1,5376
School 5	6,2	-0,66	0,4356
School 6	7,1	0,24	0,0576
School 7	7,4	0,54	0,2916
School 8	6,7	-0,16	0,0256
			<u>11.3388</u>

$$s = 1,27$$

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q3 - Q1$
 $7,65 - 6,45 = 1,2$

STANDARD DEVIATION

	Average grade chemistry	$x - \bar{x}$	$(x - \bar{x})^2$
School 1	7,4	0,54	0,2916
School 2	7,9	1,04	1,0816
School 3	4,1	-2,76	7,6176
School 4	8,1	1,24	1,5376
School 5	6,2	-0,66	0,4356
School 6	7,1	0,24	0,0576
School 7	7,4	0,54	0,2916
School 8	6,7	-0,16	0,0256
			<u>11.3388</u>

$$s = 1,27$$

MEASURES OF DISPERSION

RANGE

HIGHEST VALUE - LOWEST VALUE
 $8.1 - 4.1 = 4$

INTERQUARTILE RANGE

$Q3 - Q1$
 $7,65 - 6,45 = 1,2$

STANDARD DEVIATION

1,27

4. Construct a box plot

$$Q1 = 6.45$$

$$Q2 = \text{median} = 7.25$$

$$Q3 = 7.65$$

$$IQR = 1.21$$

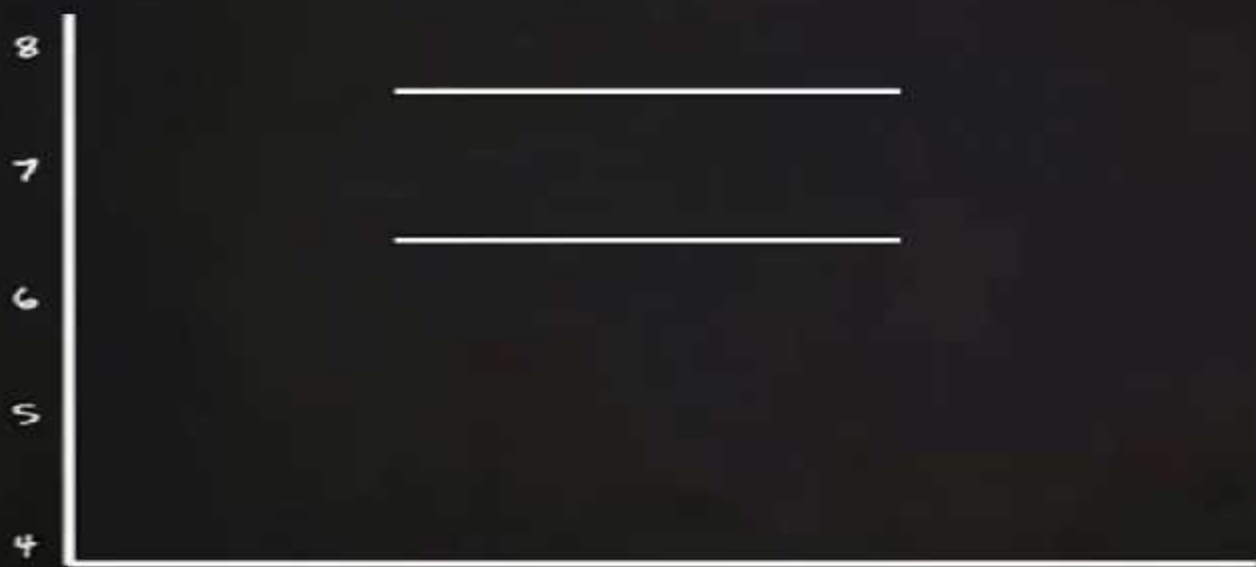
4. Construct a box plot

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$$Q3 = 7.65$$

$$IQR = 1.21$$



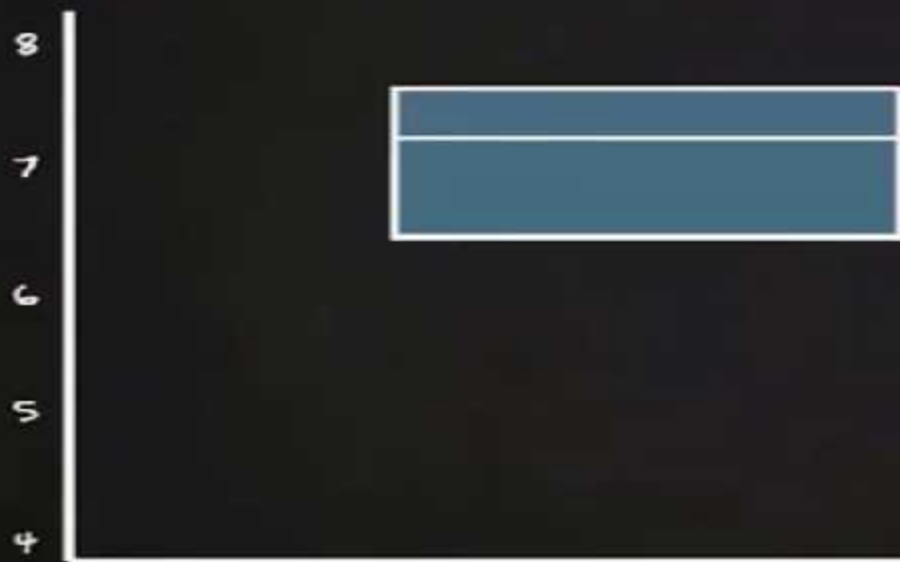
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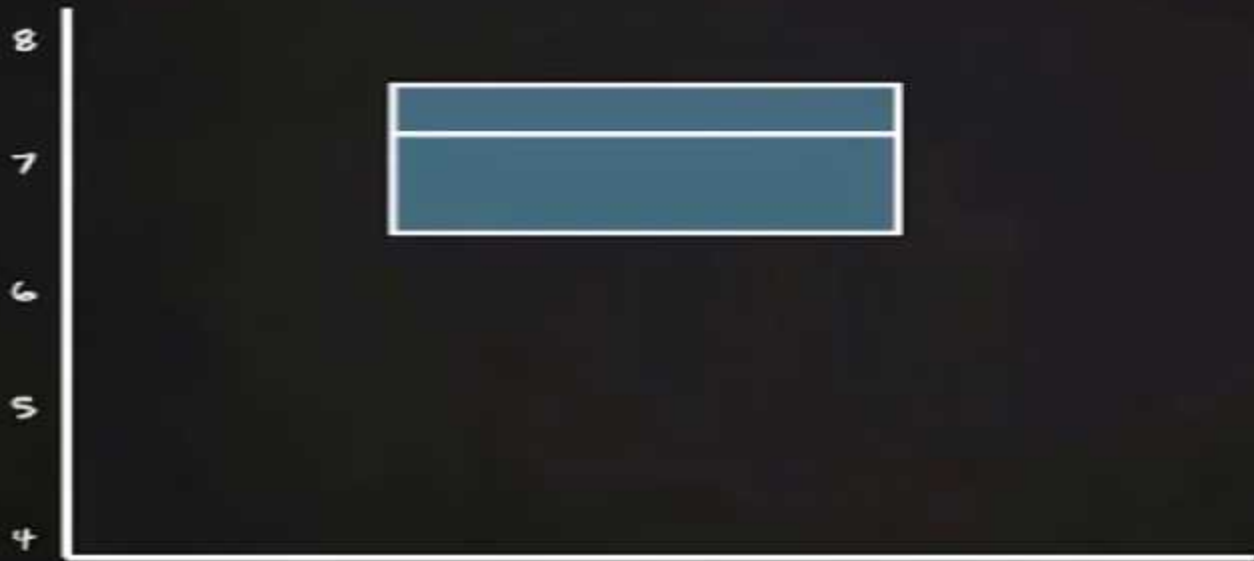
$$IQR = 1.21$$

outlier = value lower than

$$Q1 - 1.5(IQR)$$

or higher than

$$Q3 + 1.5(IQR)$$



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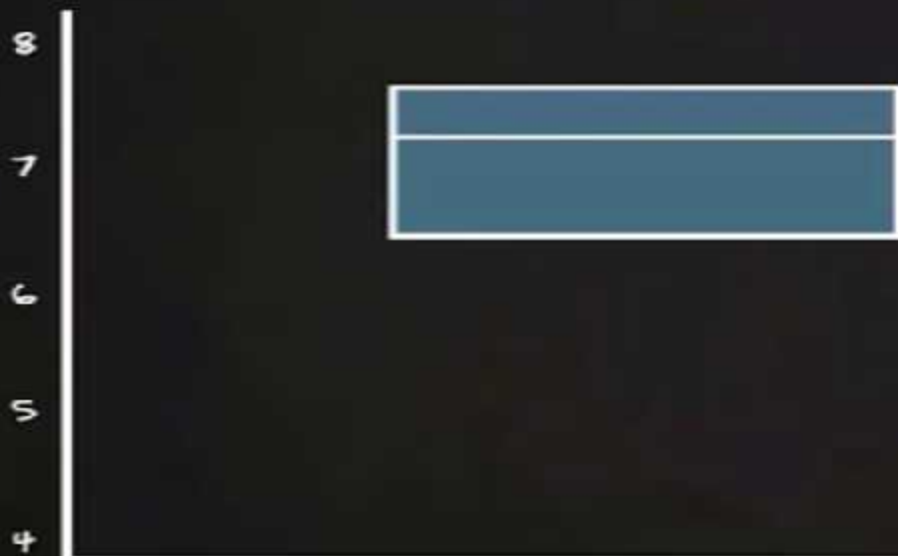
outlier = value lower than

$$Q1 - 1.5(IQR)$$

or higher than

$$Q3 + 1.5(IQR)$$

$$1.5(IQR) = 1.5 \times 1.2$$
$$= 1.8$$



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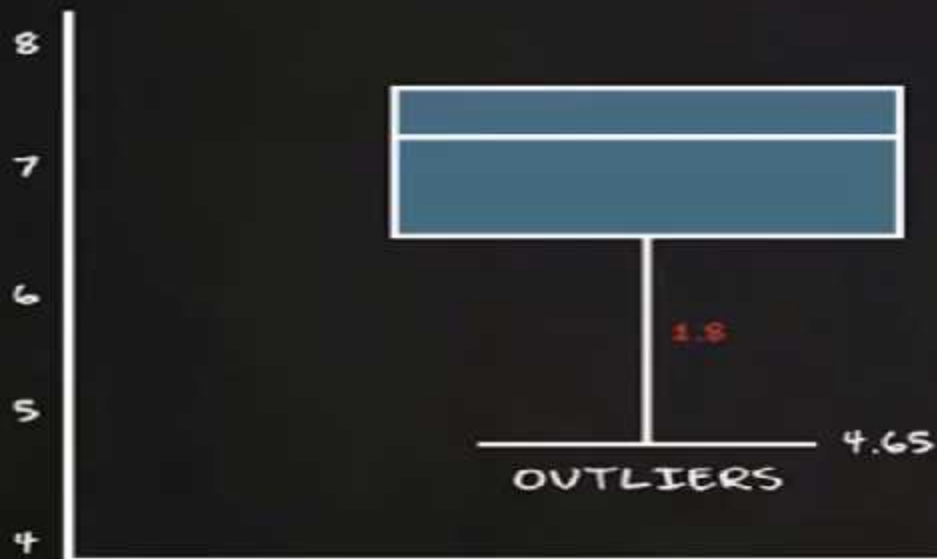
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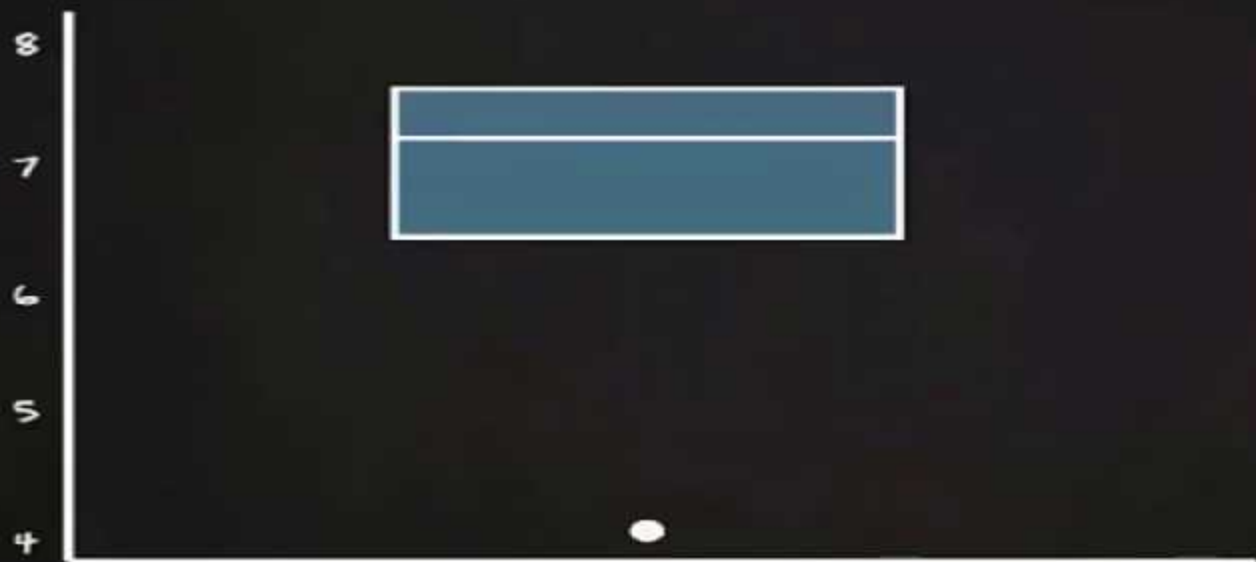
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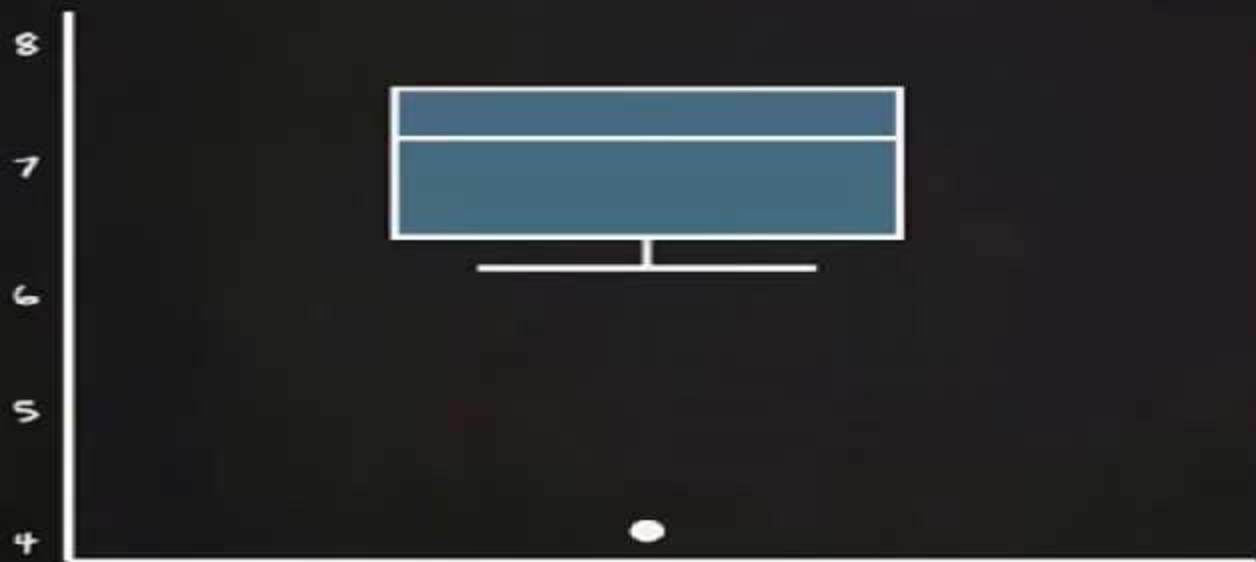
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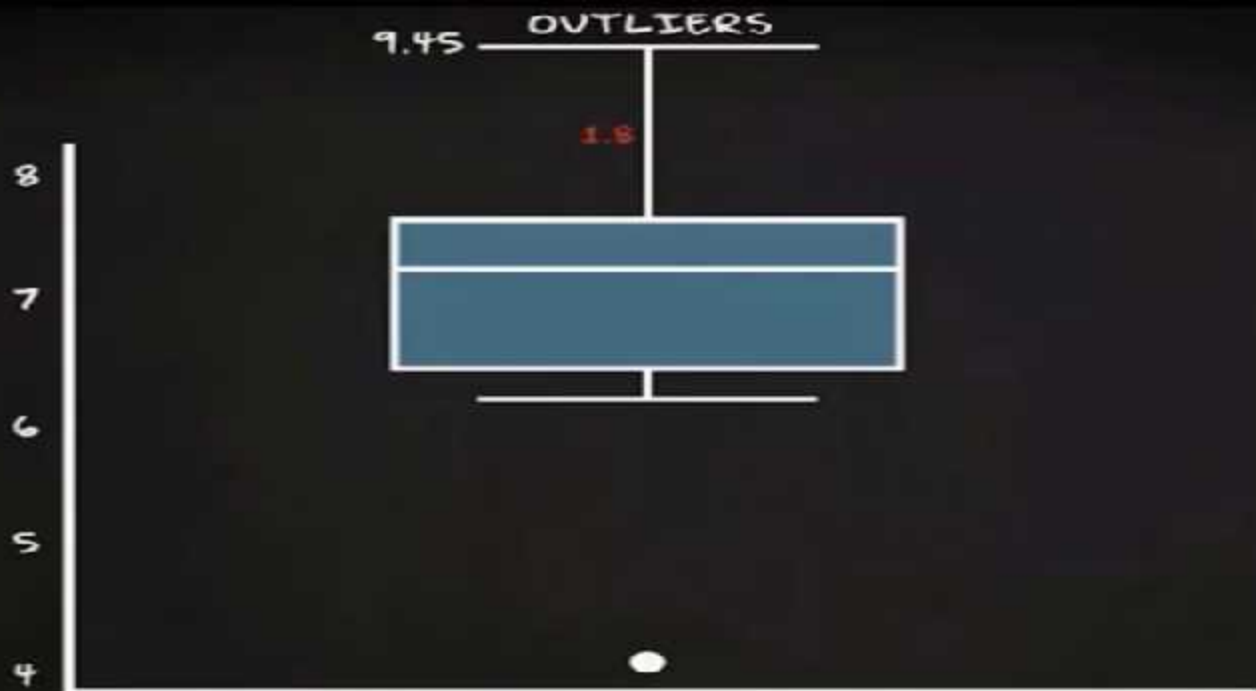
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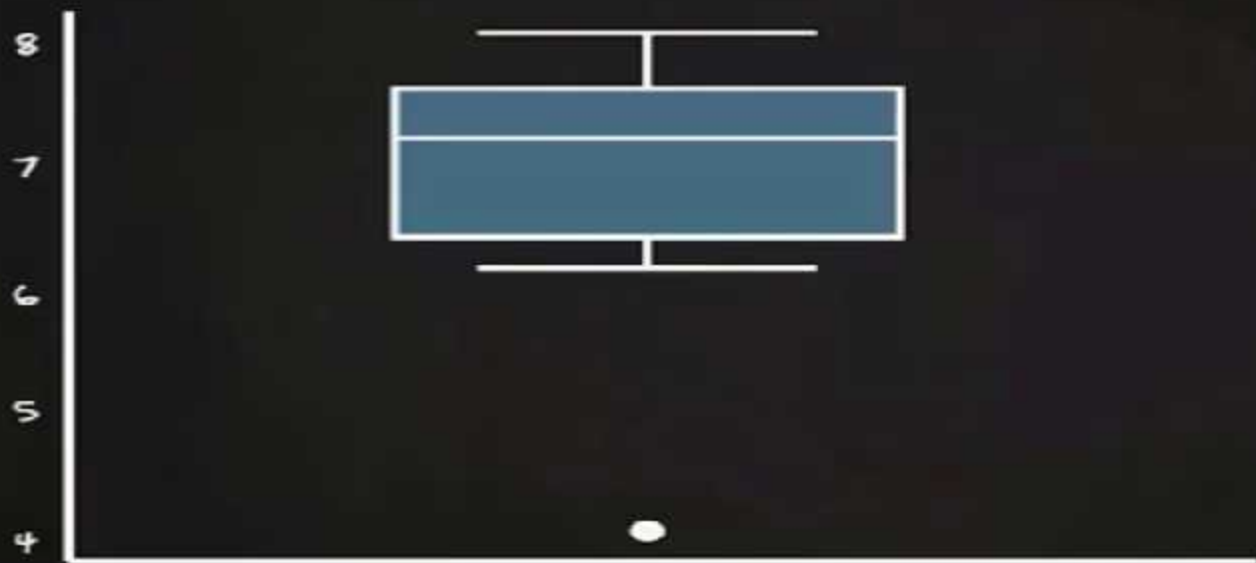
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5. What is the
z-score of school #3?

$$Z = \frac{x - \bar{x}}{s}$$

5. What is the
z-score of school #3?

$$Z = \frac{4.1 - 6.86}{1.27}$$

5. What is the
z-score of school #3?

$$Z = \frac{4.1 - 6.86}{1.27} = -2.17$$